**25901122** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) Regular EXAMINATION March 2026

## Fourth SEMESTER

## Discipline Specific Elective (DSE) - MG4DSEBOTA01 - AGRIFOOD SCIENCE AND NANOTECHNOLOGY

(2024 ADMISSION ONWARDS)

**Duration: 2 Hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Short answer

Answer any **10** questions

Each question carries **2** marks

1. Name two nanomaterials used in agriculture. [K] / [CO1]
2. List two activities involved in agrifood systems. [K] / [CO1]
3. Name two examples of physical synthesis techniques. [K] / [CO1]
4. List two bio-nanomaterials. [K] / [CO1]
5. What are food additives? Give two examples. [U] / [CO2]
6. What is the role of gold nanoparticles in nanosensor kits? [U] / [CO3]
7. What are nanosensors? [U] / [CO2]
8. What are RFID-based nanotags used in smart packaging? [U] / [CO3]
9. How do nano-fertilizers impact a crop's photosynthetic efficiency? [U] / [CO3]
10. How does nanotechnology help reduce food wastage? [U] / [CO4]

11. What is the importance of integrating nanotechnology with digital tools in modern agrifood systems? [U] / [CO4]

12. How does nanotechnology improve crop productivity? [U] / [CO3]

**[2x10 = 20]**

## Part B

Short essays

Answer any **8** questions

Each question carries **5** marks

13. Describe food products obtained from aquaculture. [K] / [CO1]

14. Describe the method of encapsulating vitamins or antioxidants using nanoliposomes. [U] / [CO2]

15. Describe the different types of nanobarcodes used in food packaging systems. [U] / [CO3]

16. Explain the benefits of nanofiltration compared to traditional filtration and thermal processing methods in fruit juice industries. [U] / [CO3]

17. Explain the procedure for preparing plant extract used in green synthesis of silver nanoparticles. [U] / [CO4]

18. How protein nanoparticles maintain the quality of packaged food? [U] / [CO2]

19. Describe how clear regulations are very important for the development of nanobased agricultural products? [U] / [CO3]

20. Write a short note on the role of IITs in nanoscience and nanotechnology research. [K] / [CO1]

21. What are three environmental issues from conventional fertilizers that traditional fertilizers are proven to lessen? [K] / [CO1]

22. Explain the role of nano-enabled "Smart Hydroponics" in urban settings. [K] / [CO1]

**[5x8 = 40]**

## Part C

Essays

Answer any **1** questions

Each question carries **10** marks

23. Outline the major applications of nanotechnology in different fields. [K] / [CO1]

24. Discuss the potential toxic effects of nanomaterials on soil health, plant systems, and human health within agricultural ecosystems. [U] / [CO3]

**[10x1 = 10]**